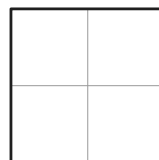
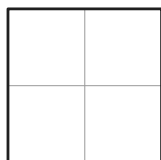
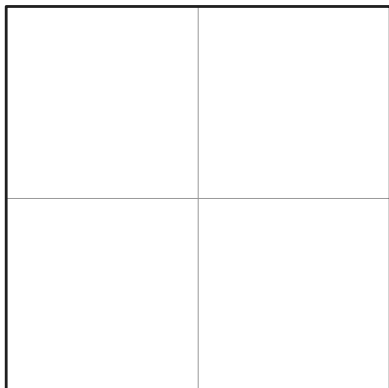
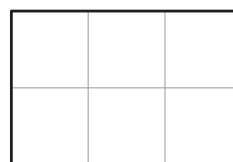
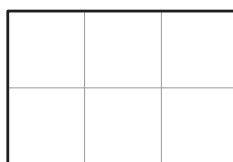
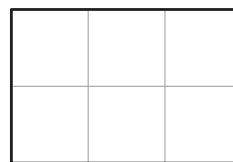
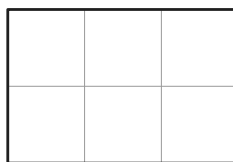
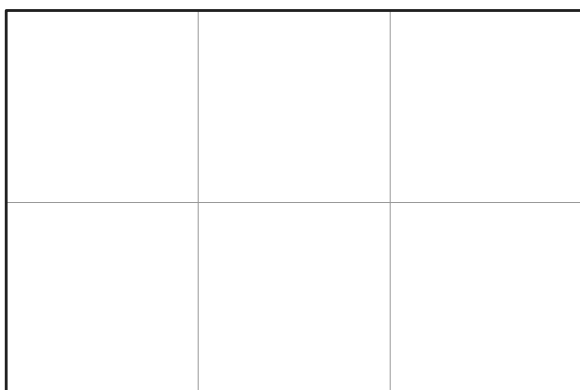


Problem 1: A domino is a paper piece that covers two squares which share a side. Cover the 2 by 2 board with dominoes: no gaps, no overlaps, nothing sticking out. Find every way. Draw each way on a small board: one thick line through the two squares of each domino. Two ways are different if some domino lies in a different place. Write how many ways in the blank.



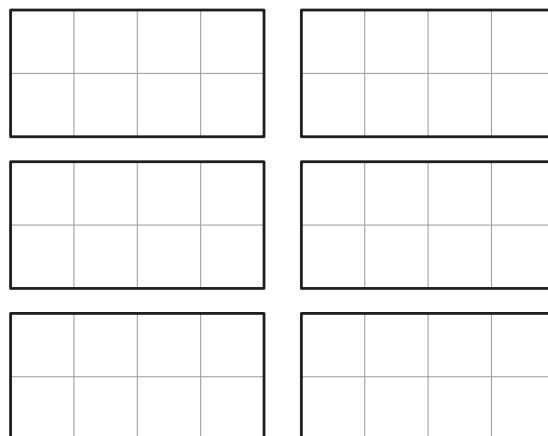
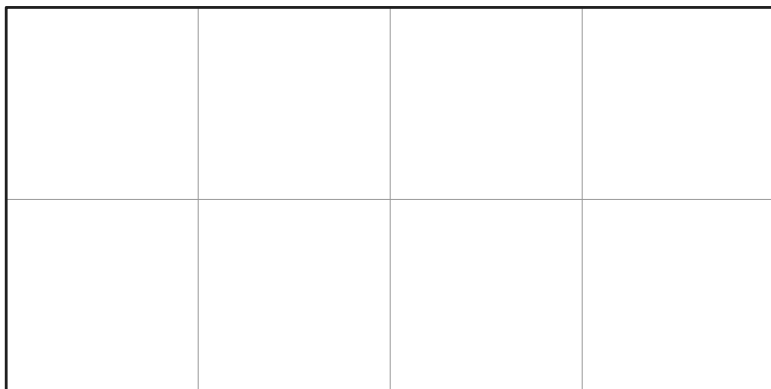
ways: _____

Problem 2: Cover the 2 by 3 board with dominoes. Find every way and draw each way on a small board. Write how many ways.



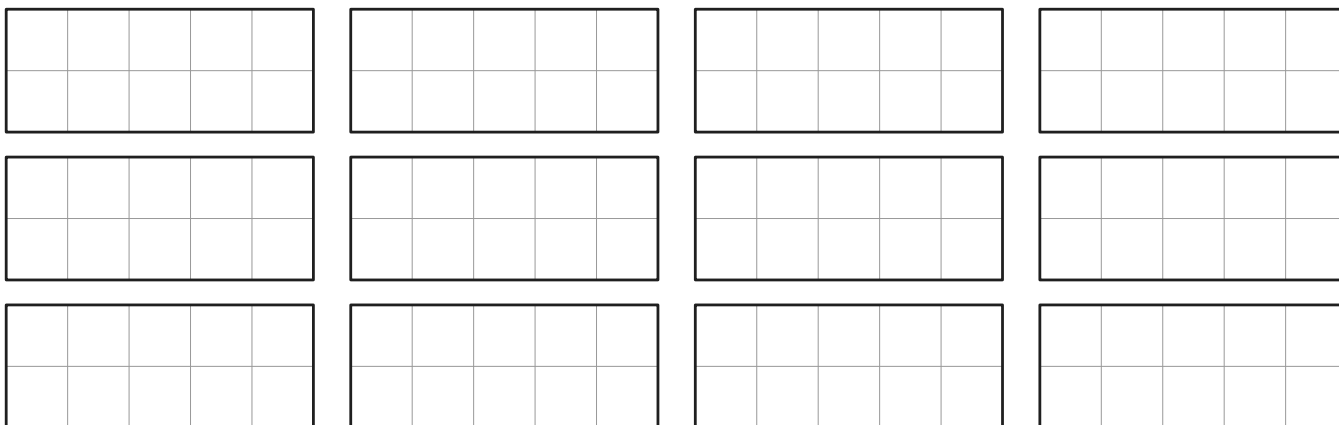
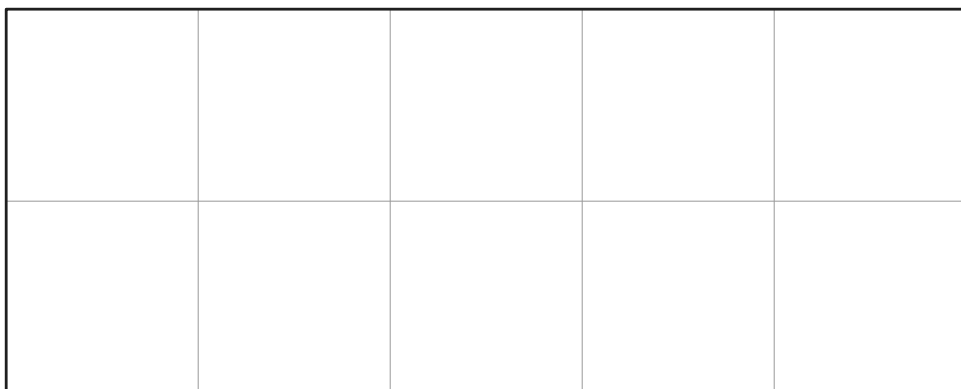
ways: _____

Problem 3: Cover the 2 by 4 board with dominoes. Find every way and draw each way on a small board. Write how many ways.



ways: _____

Problem 4: Cover the 2 by 5 board with dominoes. Find every way and draw each way on a small board. Write how many ways.



ways: _____

Problem 5: Write your counts in the table. A 2 by 1 board has 1 way. Look at the row of numbers and say to your table what you notice. Then write what you predict for the 2 by 6 board and the 2 by 7 board.

board	2 by 1	2 by 2	2 by 3	2 by 4	2 by 5	2 by 6	2 by 7
ways	1						

Problem 6: Look at your 2 by 5 boards in Problem 4. A standing domino covers two squares one above the other; a lying domino covers two squares side by side. Put a tick on every board whose left end is one standing domino. How many boards have a tick? On the boards without a tick, what covers the two squares at the left end? How many boards have no tick? Write the two numbers and add them. Then do the same with your 2 by 4 boards in Problem 3.

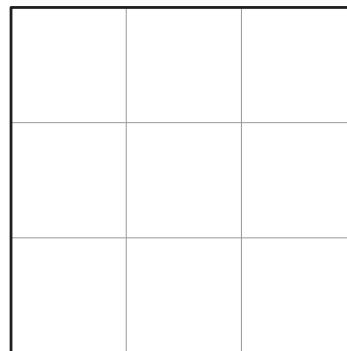
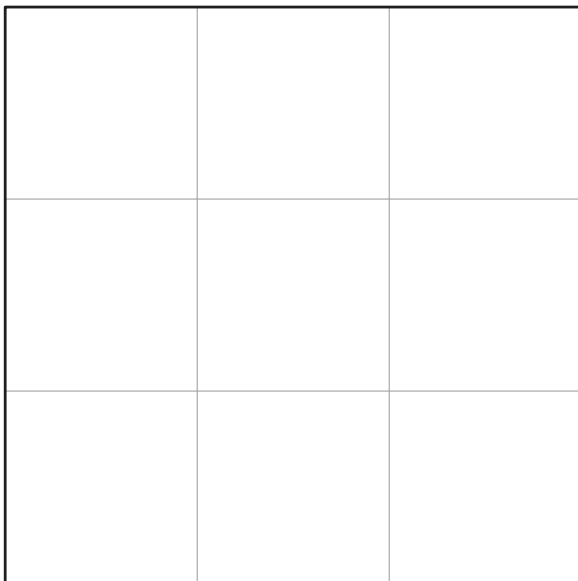
2 by 5: boards with a tick _____ boards without a tick _____ added _____ what covers the left end without a tick: _____

2 by 4: boards with a tick _____ boards without a tick _____ added _____

Problem 7: Try to cover the whole 3 by 3 board of Problem 8 with dominoes. Can you? Write yes or no, and write why in one sentence.

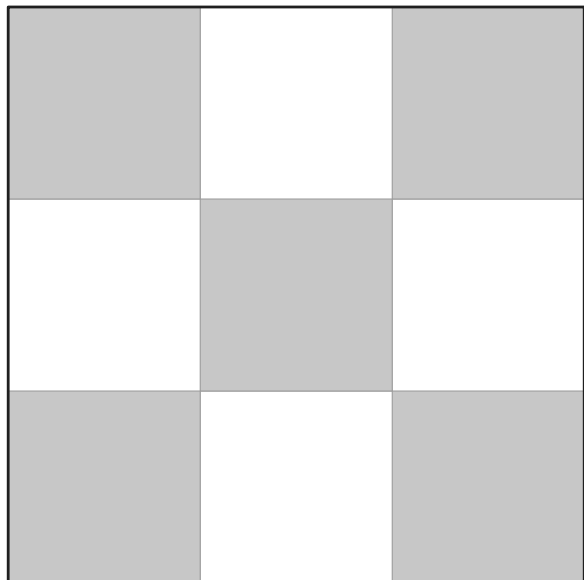
yes or no: _____ why: _____

Problem 8: Put the black paper square on one square of the 3 by 3 board. A domino may never lie on the black square. Cover the other 8 squares with 4 dominoes. Try the black square in each of the nine places. If you cannot after three different starts, write N. On the map, write Y in each square where you could cover the rest, and N where you could not.



map: Y for could, N for could not

Problem 9: This is the same board coloured like a checkerboard. Put a domino anywhere on it and look under it: how many dark squares and how many light squares does it cover? Move it somewhere else and look again. Do this five times and write the two numbers each time. Then count all the dark squares on the board and all the light squares. Now look at your map from Problem 8. Which colour are the Y squares? Which colour are the N squares?



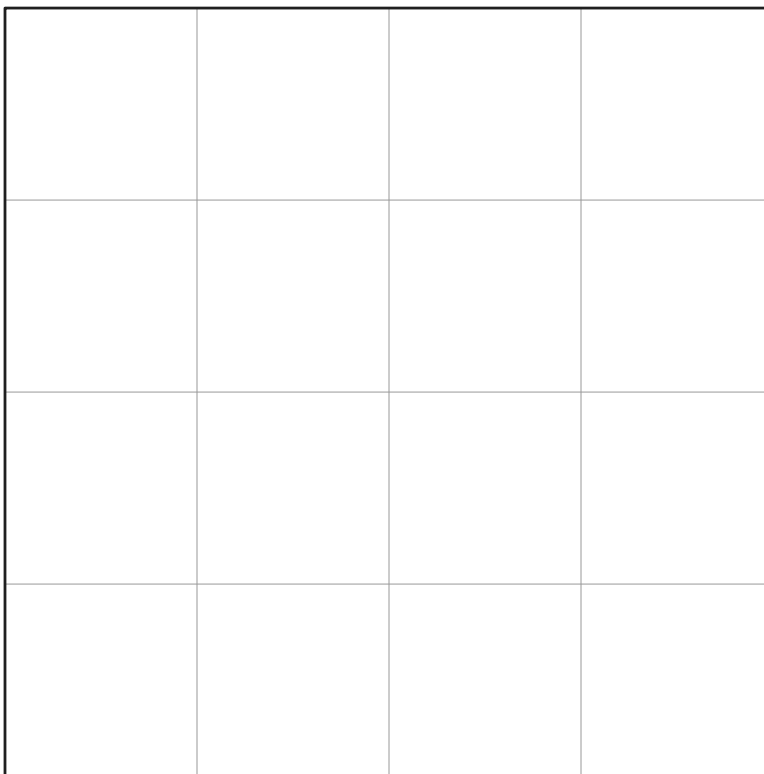
domino	dark	light
1st		
2nd		
3rd		
4th		
5th		

dark squares on the board: _____ light squares: _____

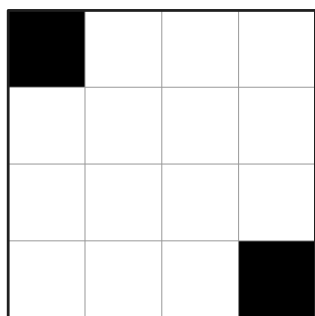
the Y squares are _____

the N squares are _____

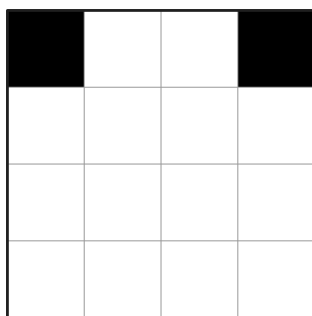
Problem 10: Put the two black paper squares on the big board where board A shows them. Cover the other 14 squares with 7 dominoes. If you can, draw the dominoes on board A. If you cannot after two different starts, write N under board A. Do the same for boards B to H.



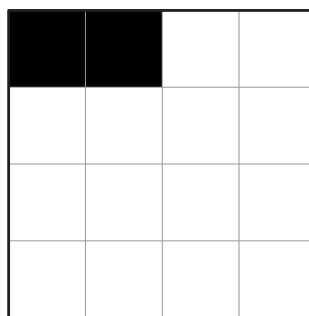
Board A



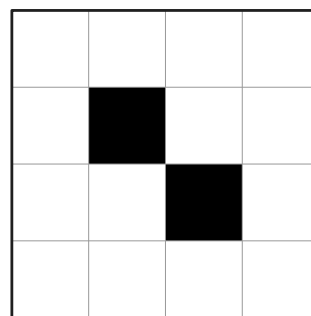
Board B



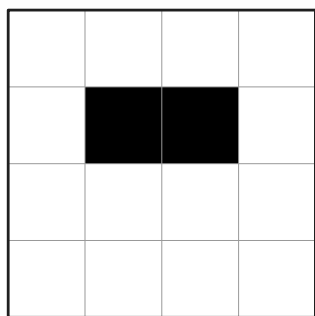
Board C



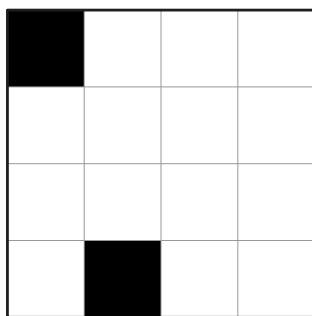
Board D



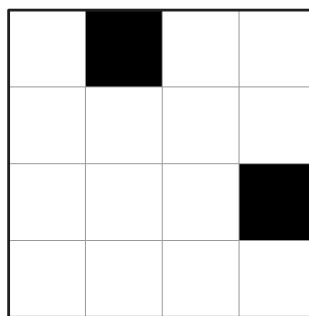
Board E



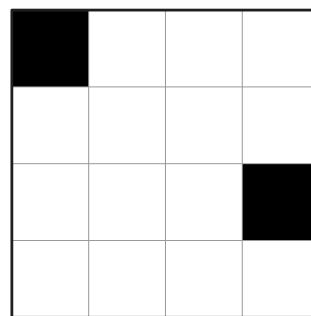
Board F



Board G

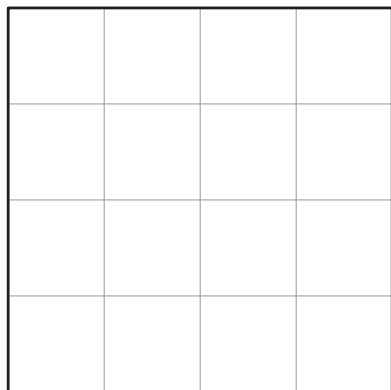


Board H

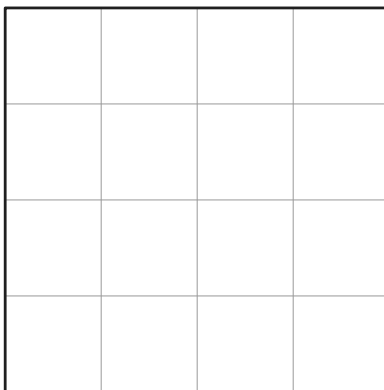


Problem 11: Choose two squares of your own for the black paper squares, a pair that boards A to H did not have. Colour them black on board J. Cover the other 14 squares with 7 dominoes; draw the dominoes, or write N under the board. Do it again on board K with another pair of your own.

Board J

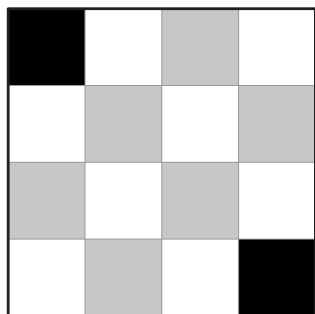


Board K

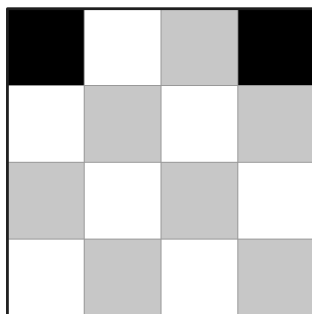


Problem 12: Here are boards A to H coloured like a checkerboard. Under each board write the colours of its two black squares, D for dark and L for light, and then copy your Y or N from Problem 10. Look at all eight. Write in one sentence what the N boards have in common, and what the Y boards have in common.

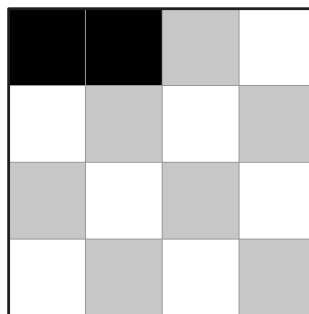
Board A



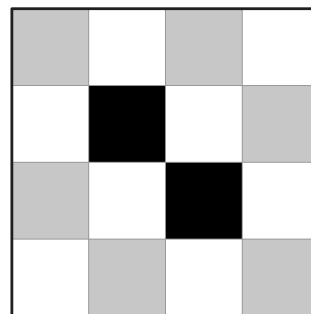
Board B



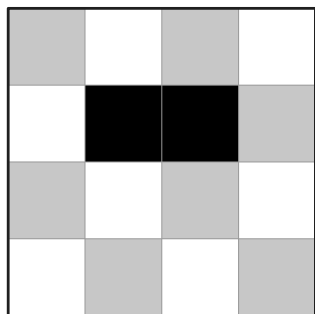
Board C



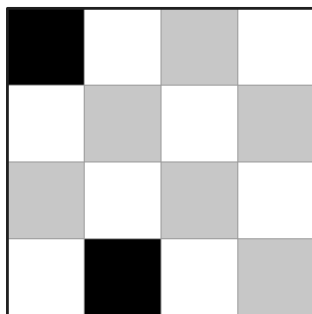
Board D



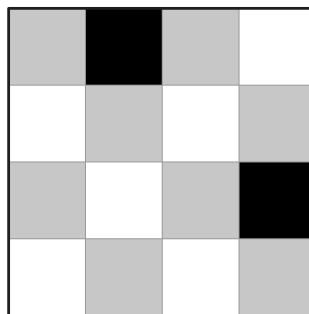
Board E



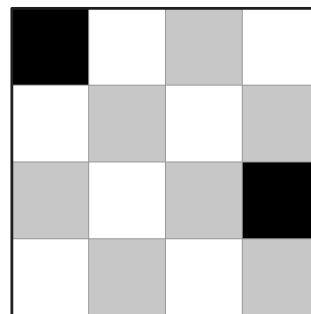
Board F



Board G



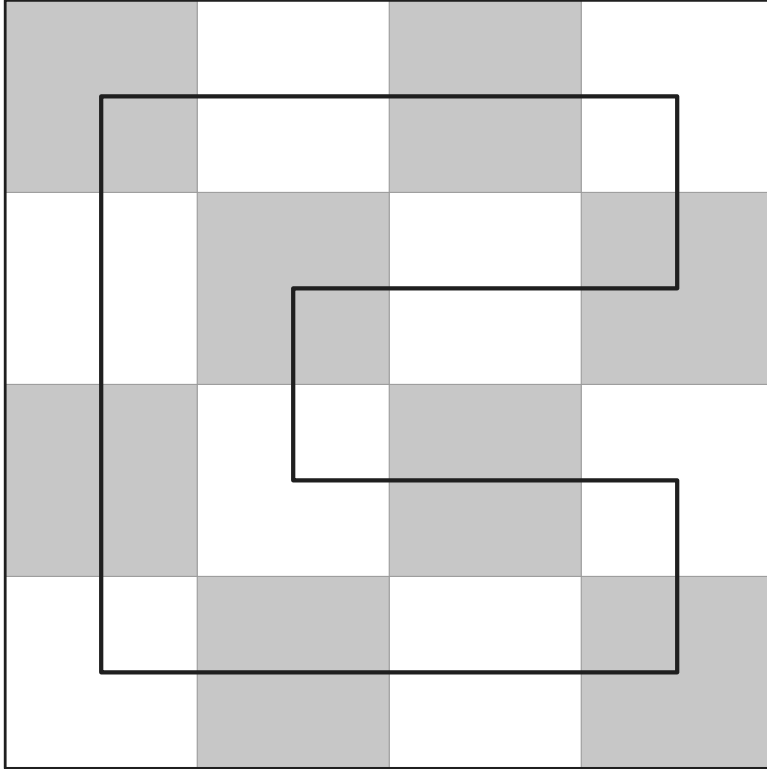
Board H



The N boards: _____

The Y boards: _____

Problem 13: On this board a path goes through all 16 squares and comes back to where it started. Choose any dark square and any light square and put the black paper squares on them. The path is now cut into two roads (a road may have no squares at all). Count the squares on each road and write the two numbers. Then lay dominoes along each road, two squares at a time, starting next to a black square. Do it again with a different dark square and light square, and a third time.



first pair:

roads of ____ and ____

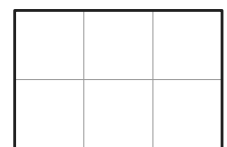
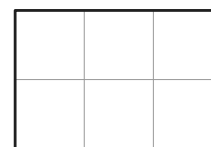
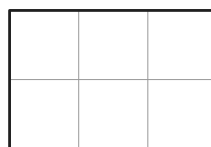
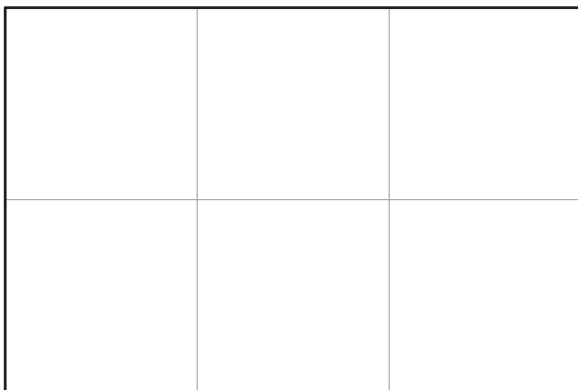
second pair:

roads of ____ and ____

third pair:

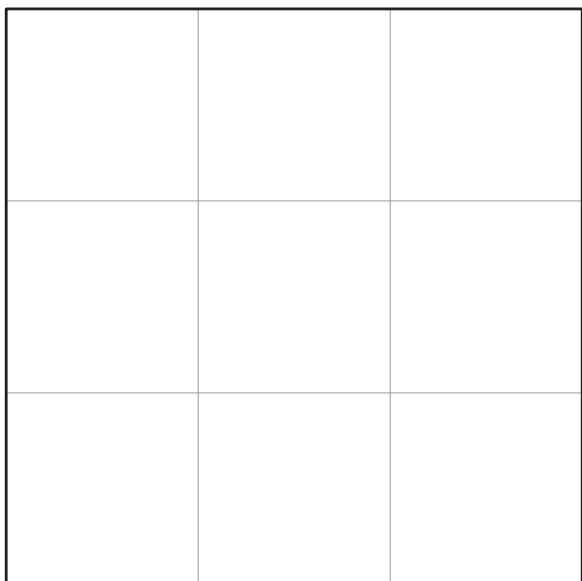
roads of ____ and ____

Problem 14: An L is a paper piece that covers three squares in an L shape. You may turn it and flip it over. Cover the 2 by 3 board with two L's. Find every way and colour each way on a small board, one colour for each L. Write how many ways.



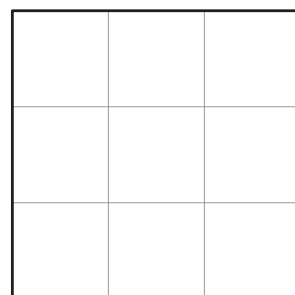
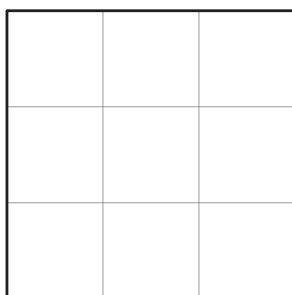
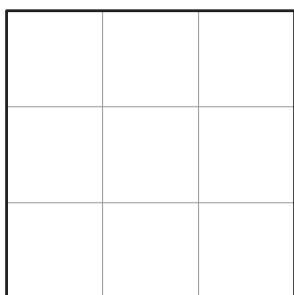
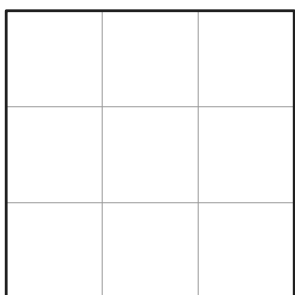
ways: _____

Problem 15: Try to cover the 3 by 3 board with three L's. Can you? Write yes or no.



yes or no: _____

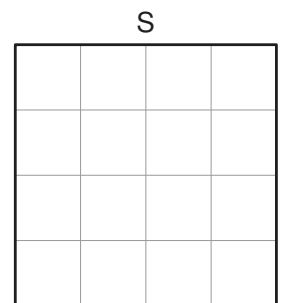
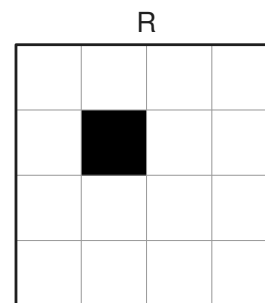
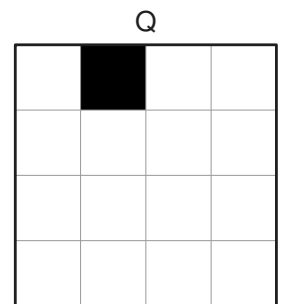
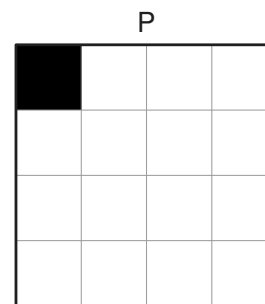
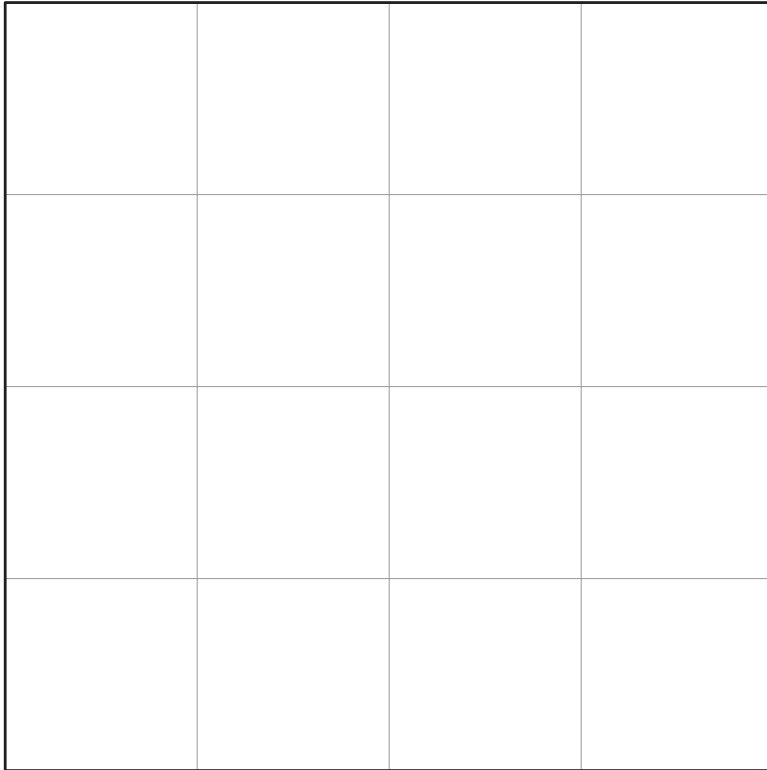
Problem 16: Put one L on the 3 by 3 board so that it covers the top-left corner square. Colour that position on a small board. Now move the L to a different position that still covers the top-left corner square, and colour it on the next small board. Find every position. Does the L ever cover two corner squares at the same time? Count the corner squares of the board and count your L's. Write what this tells you about Problem 15.



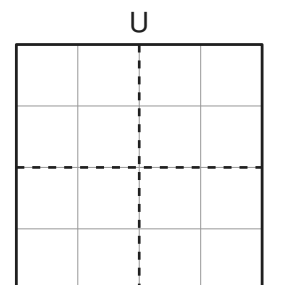
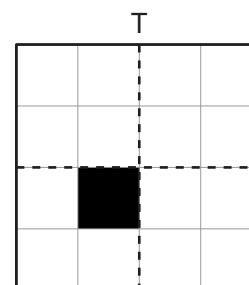
two corners at once, yes or no: _____ corner squares: _____ L's: _____

So: _____

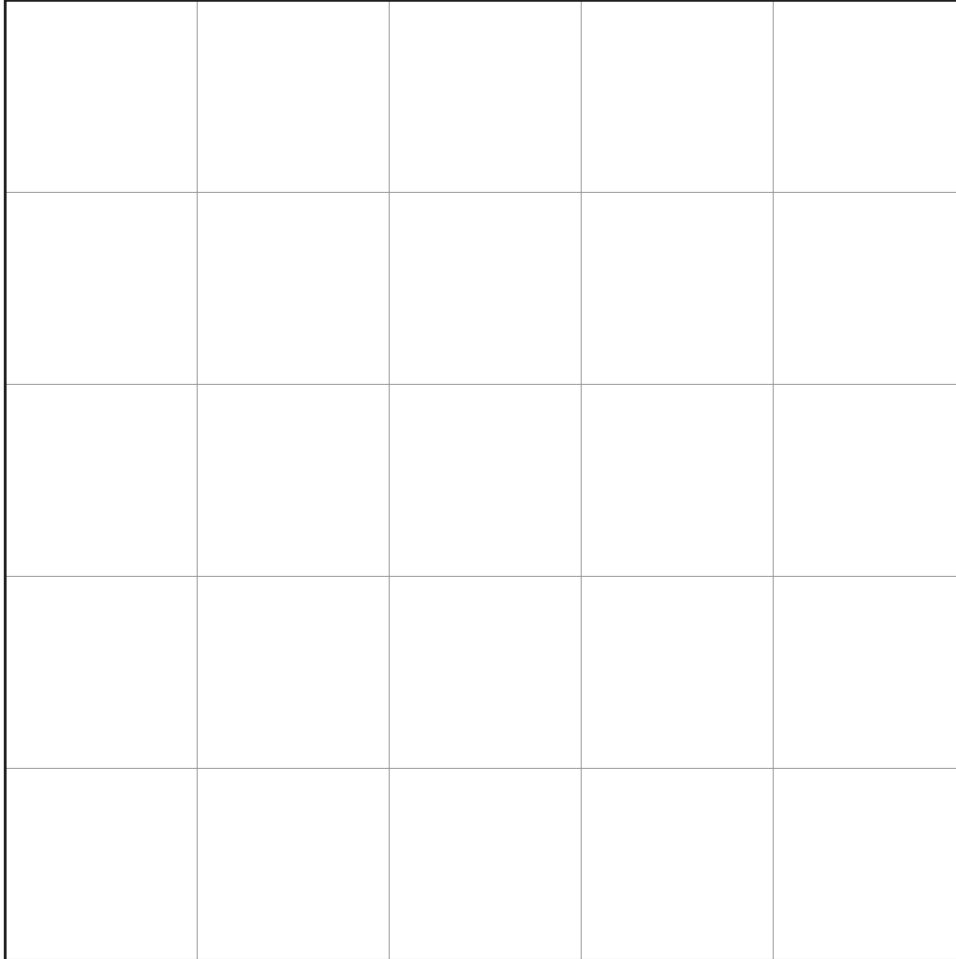
Problem 17: Put the black paper square in the top-left corner of the big board, as picture P shows. Cover the other 15 squares with five L's and colour the five L's on picture P. Then do the same with the black square at the place shown on Q, then at the place shown on R. Then choose any square you like, colour it black on S, and cover the rest.



Problem 18: A recipe for the black square anywhere: draw the two middle lines of the board, cutting it into four 2 by 2 quarters. One quarter holds the black square. Put one L in the very middle of the board, covering one square of each of the other three quarters. Now each quarter has exactly one square that is black or covered; cover its other three squares with one L. Follow the recipe on picture T and colour the five L's. Then colour a black square of your own choice on picture U and follow the recipe again.

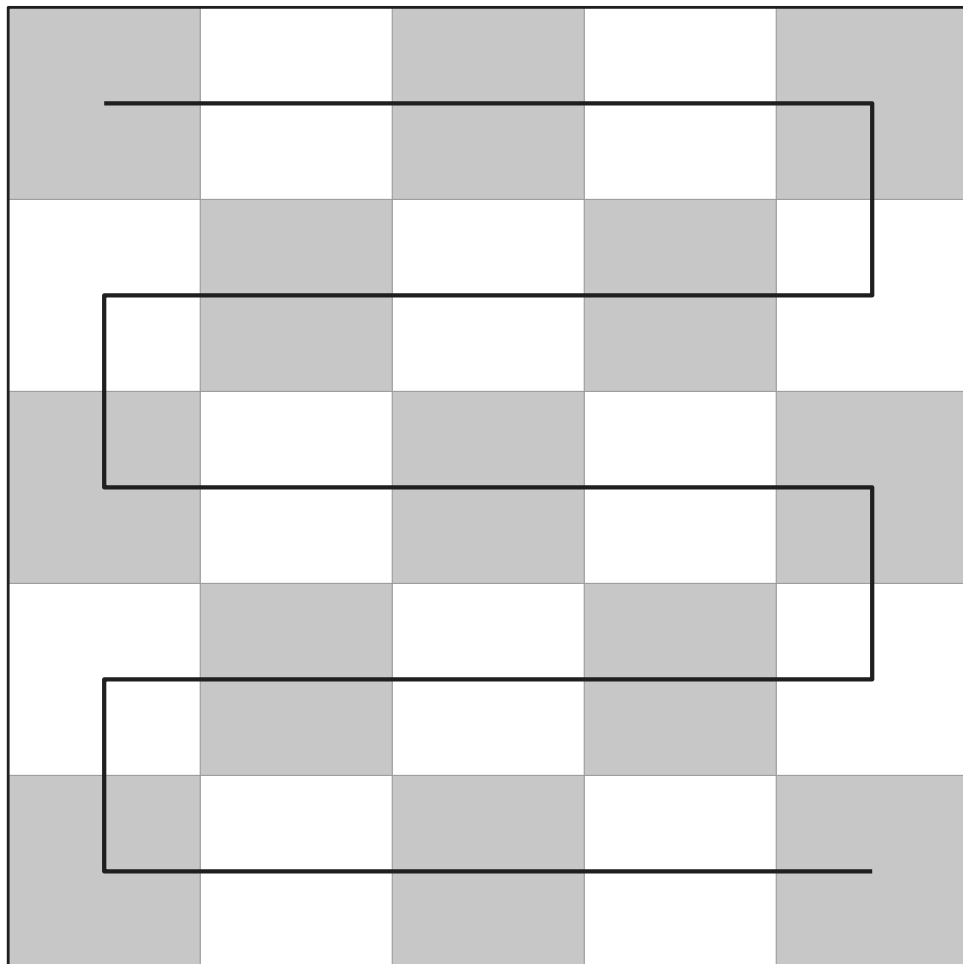


Problem 19: Put the black paper square on the 5 by 5 board at the place marked 1 on the map. Cover the other 24 squares with 12 dominoes. On the map, write Y in square 1 if you could, or N if you could not. If you cannot after three different starts, write N. Do the same for places 2, 3, 4, 5 and 6. Then fill in the rest of the map without building: the board looks the same when it is turned or flipped over, so a square that matches a tested square gets the same letter.



1	2	3		
	4	5		
		6		

Problem 20: The same 5 by 5 board, coloured like a checkerboard, with a path through all 25 squares. Count the dark squares and the light squares. Look at your map from Problem 19: which colour are the Y squares? Now put the black paper square on any dark square. The path is cut into two roads (a road may have no squares at all); count the squares on each road. Lay dominoes along each road, two squares at a time. Then put the black square on a light square and count the two roads again. Write what goes wrong.



dark squares: _____

light squares: _____

the Y squares are _____

dark black square:

roads of _____ and _____

light black square:

roads of _____ and _____

what goes wrong:

Problem 21: Use the 5 by 5 board from Problem 19 again. Put the black square at place 1 and cover the other 24 squares with eight L's. Write Y or N in square 1 of this map. If you cannot after three different starts, write N. Do places 2 to 6, then fill in the rest of the map using the turns and flips of the board. Compare this map with the domino map from Problem 19: which squares are Y on both maps, and which squares are Y on only one?

1	2	3		
	4	5		
		6		

Y on both maps: _____

Y on only one map: _____

Problem 22: This 3 by 3 board has dots on its corners. Put an L anywhere on it. How many dots can one L cover at most? Three L's cover at most how many dots? How many dots are there? Write what this says about Problem 15.

●		●
●		●

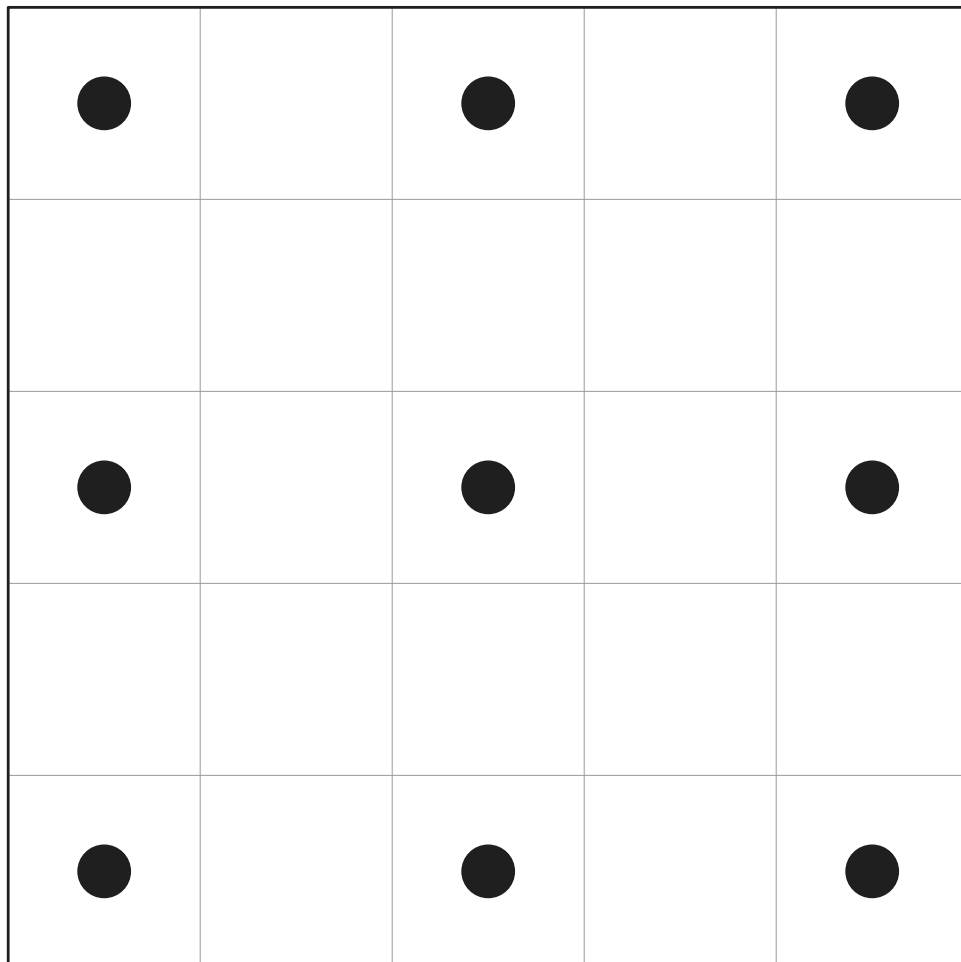
most dots one L covers: _____

three L's cover at most _____ dots

dots on the board: _____

so: _____

Problem 23: This 5 by 5 board has a dot on some squares. Draw any 2 by 2 square on the board and count the dots inside it. Draw four more 2 by 2 squares in other places and count the dots in each. Now put an L anywhere on the board: how many dots does it cover? Try ten places. Write the most dots that one L covered. Eight L's cover at most how many dots? How many dots are on the board? Write, in one sentence, what this says about where the black square must be when eight L's cover the rest.



dots in a 2 by 2 square:

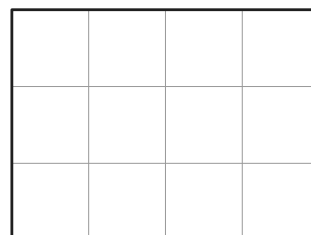
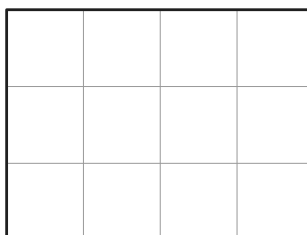
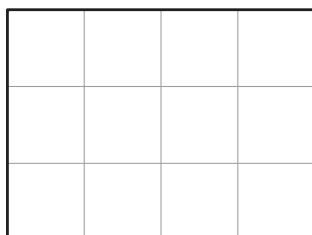
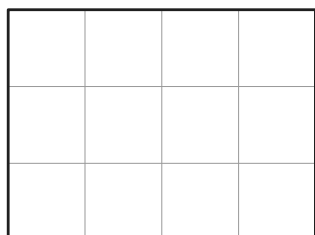
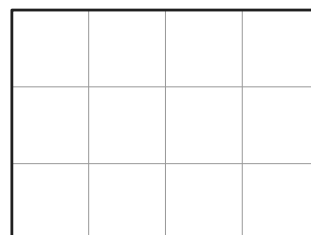
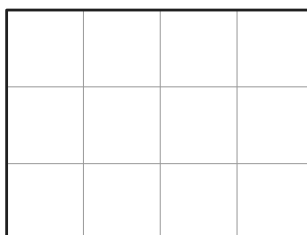
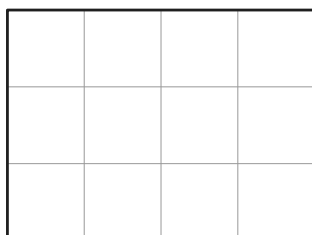
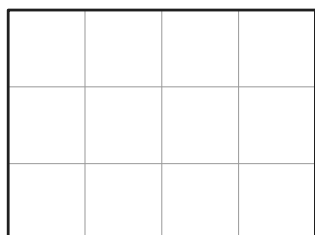
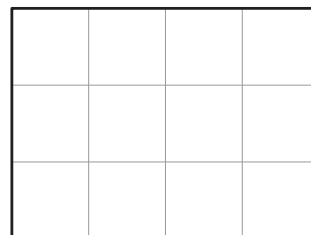
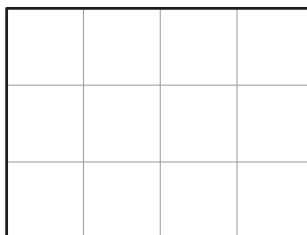
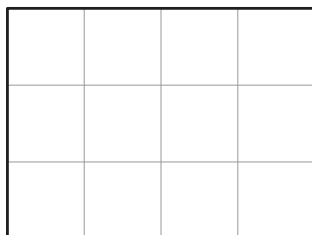
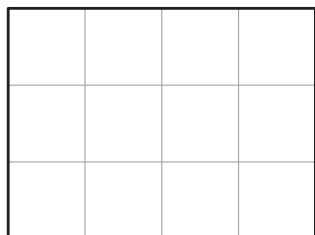
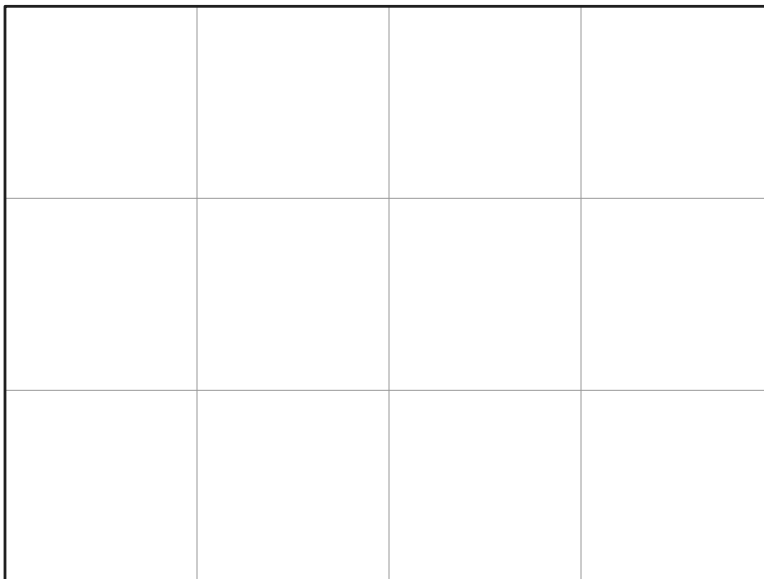
most dots under one L: _____

eight L's cover at most _____

dots on the board: _____

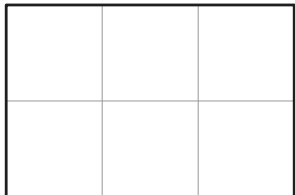
so the black square must be

Problem 24: Cover the 3 by 4 board with dominoes. Find every way and draw each one on a small board. Sort your drawings: first the ways where the top-left square is covered by a lying domino, then the ways where it is covered by a standing domino. Write how many of each kind and how many in all.

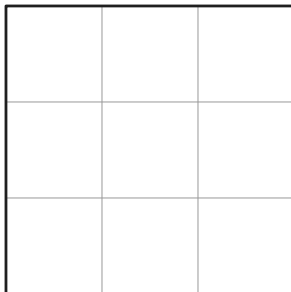


lying at top left: _____ standing at top left: _____ in all: _____

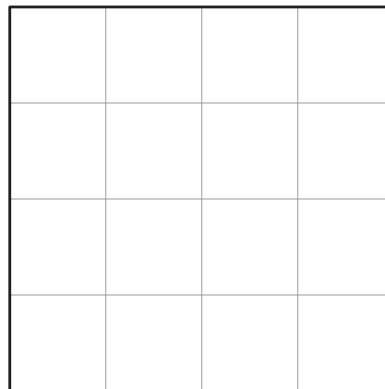
Problem 25: Use the 2 by 3 board from Problem 14, the 3 by 3 board from Problem 15 and the 4 by 4 board from Problem 17. Put dominoes on the 2 by 3 board so that no more dominoes can be added anywhere, using as few dominoes as you can. Draw your best answer on the small 2 by 3 board and write the number of dominoes. Then do the same on the 3 by 3 board and on the 4 by 4 board: fewest dominoes so that no more fit.



fewest: _____

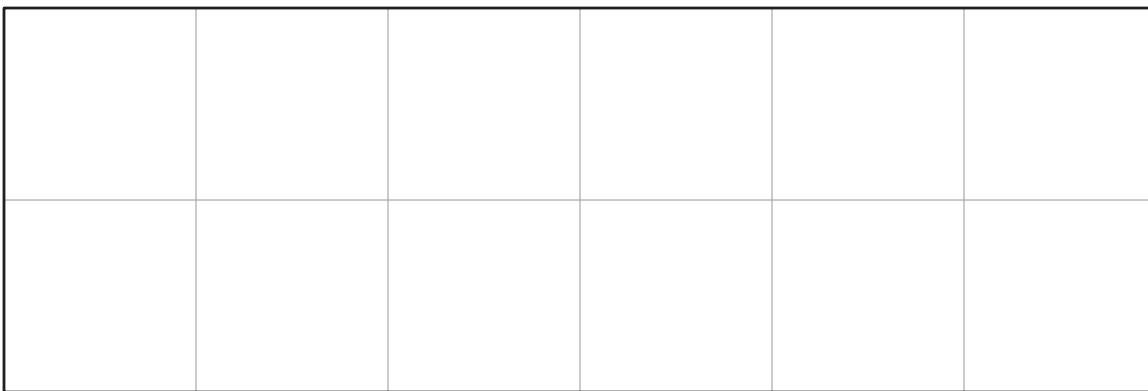


fewest: _____

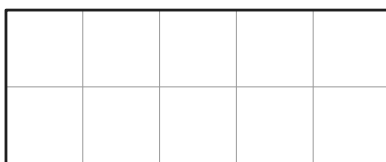


fewest: _____

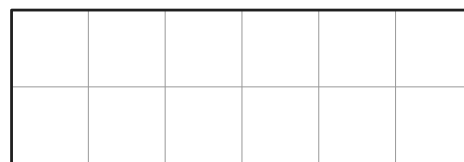
Problem 26: Fewest dominoes so that no more fit, on the long boards: the 2 by 4 board from Problem 3, the 2 by 5 board from Problem 4 and the 2 by 6 board below. Draw your best answer for each and write the number. Then do the 2 by 2 board from Problem 1 and write all five numbers in the row. Put your best 2 by 3 and your best 2 by 6 side by side. What do you notice? Without building, predict the fewest for the 2 by 9 board.



fewest: _____



fewest: _____



fewest: _____

2 by 2: _____ 2 by 3: _____ 2 by 4: _____ 2 by 5: _____ 2 by 6: _____

what I notice: _____

2 by 9: _____ dominoes